

Implemented Positional Encoding in Training_new

Notation. Let $G = (\mathcal{V}, \mathcal{E})$ be the node graph used for PE. For node $i \in \mathcal{V}$:

$$\begin{aligned} d_i &: \text{node degree,} \\ s_i \in \{0, 1\} &: \text{slack indicator,} \\ g_i \in \{0, 1\} &: \text{ground indicator,} \\ \mathcal{S} = \{u \in \mathcal{V} : s_u = 1\} &: \text{set of slack nodes.} \end{aligned}$$

BFS distance. BFS (Breadth-First Search) gives shortest path length in *number of edges*. Thus

$$h_i = \min_{u \in \mathcal{S}} \text{dist}_{\text{BFS}}(i, u), \quad h_i \leftarrow \min(h_i, H),$$

where $H = \text{DK_PE_HOPCAP}$.

The node positional encoding is

$$\mathbf{p}_i = \begin{bmatrix} \frac{\log(1 + d_i)}{h_i} \\ \frac{1}{\max(10, H/3)} \\ s_i \\ g_i \\ \frac{d_i}{d_{\max}} \\ \frac{\log(1 + d_i)}{\log(1 + d_{\max}) + 10^{-6}} \end{bmatrix} \in \mathbb{R}^6, \quad d_{\max} = \max_{j \in \mathcal{V}} d_j.$$

Why log terms. The transforms $\log(1 + d_i)$ and $\log(1 + d_{\max})$ compress heavy-tailed degree scale, so high-degree nodes do not dominate the feature magnitude. Using $\log(1 + x)$ is also well-defined at $x = 0$ and keeps features numerically stable.

Hence,

$$\text{PE_DIM} = 6.$$

Per-node encoder input is

$$\mathbf{x}_i^{\text{node}} = [\mathbf{V}_i^{\text{init}}, \Delta \mathbf{V}_i \odot m_i, m_i, \mathbf{p}_i],$$

where

$$\begin{aligned} \mathbf{V}_i^{\text{init}} \in \mathbb{R}^2 &: \text{initial voltage (real, imag),} \\ \Delta \mathbf{V}_i \in \mathbb{R}^2 &: \text{voltage residual target,} \\ m_i \in \{0, 1\} &: \text{voltage visibility mask,} \\ \odot &: \text{elementwise product,} \\ [\cdot, \cdot, \cdot] &: \text{feature concatenation.} \end{aligned}$$

No-PE ablation:

$$\mathbf{p}_i \leftarrow \mathbf{0}.$$